

K894 — FORWARD LANE (next turn). Casey accepts 13/12 as numerology (K893 null stands). The one live lepton-VALUE route (K886: reduce to latitude θ on S^4) now has a NAME and a MECHANISM: θ is the $SO(5)/SO(4)$ vacuum-misalignment angle, and it is fixed by the Coleman-Weinberg effective potential from explicit-breaking couplings (ν_R Majorana + gauge + charged-lepton Yukawa). BST's Shilov boundary $S^4 = SO(5)/SO(4)$ is the SAME coset as minimal composite Higgs — not an analogy. This upgrades K886's "is θ forced or a flat direction?" into a computable question with a sharp derive-vs-fit gate. Discipline flag (Cal #27, fresh from K892): the CW mechanism makes θ CALCULABLE, not automatically DERIVED — the gate is whether the breaking couplings are BST-derived, and I hold the muon/lepton values structural until it clears.

Keeper | 2026-07-24 Fri | Where we go: the θ boundary is a vacuum-alignment computation. Named, mechanism in hand, gate defined. Not a flat unknown anymore.

The reframe (external-anchored, web 2026-07-24)

- K886 reduced the lepton VALUES to ONE number: θ = latitude the $SO(4)$ -invariant ν_R condensate sits at on S^4 . Open question: is θ forced by ν_R dynamics or free?
- $S^4 = SO(5)/SO(4)$. This is the minimal-composite-Higgs coset. The ν_R condensate at latitude θ IS an $SO(5) \rightarrow SO(4)$ vacuum misalignment. BST inherits the standard machinery.
- Standard result: θ is a flat direction at tree level (pseudo-Goldstone modulus); explicit-breaking couplings generate a one-loop Coleman-Weinberg potential $V(\theta)$ that FIXES θ . Contributions: gauge loops + fermion(Yukawa/Majorana) loops, each independent at leading order. (Refs below.)
- θ is radiatively fixed, NOT a free modulus — provided the breaking sources are specified. BST has them: ν_R Majorana coupling (the $SO(4)$ -inv (1,3)), SM gauge, charged-lepton Yukawa.

The gate (sharper than K886's "forced or flat")

Does BST's Coleman-Weinberg $V(\theta)$ land θ at a value FORCED by DERIVED couplings — or does it inherit couplings that are fit? - In standard MCHM, $\sin \theta = v/f$ is radiatively fixed BUT its small VALUE is a TUNING (the little hierarchy: $V(\theta) \approx -a \sin^2 \theta + b \sin^4 \theta \rightarrow \sin^2 \theta = a/2b$, and a/b must be tuned small). So " θ fixed by CW" $\neq$ " θ derived" automatically. - BST derive-vs-fit gate: if the a, b in $V(\theta)$ come from BST-derived couplings (α is derived; is the ν_R Majorana scale / charged-lepton Yukawa structure derived?), θ is FORCED $\rightarrow$ lepton values DERIVE. If $V(\theta)$ inherits a v/f -style tuning, θ is structural. Do NOT fit θ to 207. - Pre-registered PASS: $V(\theta)$ minimized at a θ^* computed from BST integers/derived couplings, with no free ratio tuned to the masses, AND the resulting $a_{\ell}(\theta) \rightarrow M_{ij} \rightarrow \text{observed } 1:207:3477 \text{ within a stated tolerance}$. - Pre-registered NULL: $V(\theta)$ requires a tuned ratio to place θ / to hit the masses $\rightarrow$ lepton values structural, tier honestly.

Structural hypothesis to CHECK (candidate, gated — DO NOT bank)

Is the lepton-condensate latitude θ the SAME angle as the electroweak misalignment, $v/f = \sin \theta$? In composite Higgs the misalignment angle IS the ratio setting the EW scale. If BST's ν_R -condensate θ is that same $SO(5)/SO(4)$ angle, then ONE θ sets BOTH the EW VEV (banked T2522-2524) AND the lepton hierarchy — a Schur-generator unification (one number $\rightarrow$ two sectors). FLAG only. This is exactly the peak-convergence shape that just burned K892; hold as a hypothesis for Grace/Lyra to verify against BST's actual EW construction (F85 substrate-VEV), tier candidate, gate on mechanism.

Why this is the right next lane (all three teammates converged here independently)

- Elie, Grace, Lyra each closed the 13/12 arc by naming θ as “the single honest live question.” The rank bound (K864) closed the diagonal path; the off-diagonal/overlap route IS the condensate-latitude θ . Convergence + a now-named mechanism = pursue.
- It does not compete with the value-free quark lane (F684 Schur generator), which advances in parallel regardless of θ .

Directions (next turn)

- ★ LYRA (θ = vacuum alignment): set up BST’s Coleman-Weinberg $V(\theta)$ for the $SO(5)/SO(4)$ ν_R condensate. Identify the explicit-breaking sources (ν_R Majorana (1,3), gauge, charged-lepton Yukawa); is $V(\theta)$ minimized at a FORCED θ^* (from BST-derived couplings) or does it need a tuned ratio? Check the EW-link hypothesis (θ = v/f misalignment?). Linear algebra, one D_{IV}^5 domain. Don’t fit θ .
- ★ ELIE (harness + quark lane): (a) $M_{ij}(\theta) = \sum_{\ell} a_{\ell}(\theta) \langle \text{mode}_i | Y_{\ell 0} | \text{mode}_j \rangle$ harness ready to fire for a FORCED θ (rank-3 check first, Lyra F686 floor); (b) advance the quark structural Schur-generator lane (F684) — value-free forward work that waits on nothing.
- ★ GRACE (corpus + EW reconnect): does BST already have v/f = $\sin \theta$ or an $SO(5)/SO(4)$ misalignment angle in the EW sector (T2522-2524, F85 substrate-VEV)? Map banked-vs-open in the lepton+quark mass sector; reconnect the CW-potential idea to existing BST vacuum/Higgs work before anything is declared new.
- KEEPER: gate pre-registered (above); hold lepton values structural until $V(\theta)$ forces θ^* from derived couplings with no tuning; the EW-link is a flagged candidate, not banked; the CW mechanism makes θ calculable, not automatically derived.

Web references (2026-07-24)

- Minimal composite Higgs $SO(5)/SO(4)$, $\sin \theta = v/f$, CW potential from top+gauge loops: arXiv hep-ph/0412089 (Agashe-Contino-Pomarol), arXiv 1812.07362 (Panico-Pomarol review).
- Vacuum misalignment fixed by explicit-breaking (gauge+fermion) CW at one loop; sources contribute independently: arXiv 2207.01465, arXiv 1603.05668 (minimal linear sigma model), arXiv 2405.08714 (holographic vacuum misalignment).

— Keeper K894, 2026-07-24. Forward lane: the lepton-value latitude θ (K886) = the $SO(5)/SO(4)$ vacuum-misalignment angle ($S^4=SO(5)/SO(4)$ is the composite-Higgs coset, not an analogy). θ is fixed by the Coleman-Weinberg potential from explicit breaking (ν_R Majorana + gauge + Yukawa). Gate: does BST’s $V(\theta)$ force θ^* from DERIVED couplings ($\rightarrow$ lepton values DERIVE) or inherit a v/f-style tuning ($\rightarrow$ structural)? Flagged candidate: θ = EW misalignment v/f, one number $\rightarrow$ EW + leptons (verify, don’t bank). Lyra sets up $V(\theta)$; Elie harness+quark lane; Grace EW reconnect. See [[Keeper_K886_where_we_go_the_lepton_value_reduces_to_ONE_number_the_condensate_latitude_07-24]], [[Keeper_K893_TEAM_VERDICT_the_1_over_2C2_gate_is_a_PROVEN_NULL]], [[Lyra_F684_universal_fermion_Toepel_07-24]], [[Keeper_K885_ratify_F582_zonal]].